Quiz # 2 (Data Structures) CLO2 **BS DS Fall 2025 Afternoon**

Name:_____ **Roll number:_____**

Time Allowed: 20 min **Total Marks: 30**

Q1. Write asymptotically equal algorithms in a single line: [10 Marks]

$3n^3 - 25n + 16$, $\frac{5n^2+500\,n+16}{2n}$, $100n^2lg_3n + 2500$, $13n^2 - 100006$, $n^3 + n^2lgn$, 2n-250, $\sqrt[2]{n^{\frac{2}{3}}} + 25$

1. $3n^3 - 25n + 16$, $n^3 + n^2lgn$
2. $\frac{5n^2+500\,n+16}{2n}$, 2n-250
3. $100n^2lg_3n + 2500$
4. $13n^2 - 100006$
5. $\sqrt[2]{n^{\frac{2}{3}}} + 25$

Q2. Find time equation and asymptotic running time: [10+10= 20 Marks]

```
for (i = 1; i <= 500 ; i++)          501 (500 times true and 1 time false)
   for (j = n; j >=1 ; j = j / 3)    500 (log3 n + 1)
          a = i + j;                 500 (log3 n)
```

T(n) = 501 +500 **$\log_3$ n** + 500 + 500 **$\log_3$ n**

T(n) = 1000 **$\log_3$ n** + 1001

Asymptotic Time: 1000 $\log_3$ n = O($\log_3$ n)

```
i=1
while (i<=n)                         n+1 (n times true and 1 time false)
    j=n                              n
    while (j>=i)                     ∑_{i=1}^{n} n − i + 2
          a=i+j                      ∑_{i=1}^{n} n − i + 1
          j - -                      ∑_{i=1}^{n} n − i + 1
    i + +                            n
```

T(n) = n + 1 + $\sum_{i=1}^{n} n - i + 2$ + $\sum_{i=1}^{n} n - i + 1$ + $\sum_{i=1}^{n} n - i + 1$ + n

T(n) = n + 1 + $\sum_{i=1}^{n} n - i + 2$ + $\sum_{i=1}^{n} n - i + 1$

T(n) = n + 1 + $\frac{(n+1)(n+2)}{2} - 1$ + $\frac{(n)(n+1)}{2}$ + $\frac{(n)(n+1)}{2}$ + n

T(n) = An^2 + Bn+ C